

CampusMind: Problem Definition

ICS 2413 Artificial Intelligence. Part 1 (Week 1). Owen Monda

A. Problem statement

A new student stands at one place on campus and needs to get to another. They cannot see the whole campus, so they need directions. The campus is six locations joined by walking paths, and every path has a length in kilometres. CampusMind has to choose a chain of paths that starts where the student is and ends where the student wants to be. Any chain that connects the two places is a valid answer, but the useful answer is the chain with the smallest total distance.

To a person this is trivial, because they look at the map and see it. A program cannot look. It needs the map turned into a fixed set of ingredients: where can I be, where do I start, where do I want to end, what can I do next, and what does each move cost. The rest of this document defines those ingredients.

The six locations

Code	Location	Purpose
L1	Main Gate	Primary entrance to campus
L2	Administration Block	Admissions, registration, finance, general admin
L3	Library	Academic learning and study resources
L4	Science Laboratory	Science and computing practical facilities
L5	Student Affairs	Student welfare, guidance, related services
L6	Cafeteria	Food and refreshment services

B. Initial state

The initial state is the location where the student is when they ask for help. For the run I will use throughout Weeks 1 to 3, it is `CurrentLocation = Main Gate (L1)`, because that is where every new student arrives.

C. Goal state

The goal state is the location the student wants to reach. For my selected route it is `CurrentLocation = Science Lab (L4)`. The problem is written so that any of the six locations can be the start or the goal. Only the pair I pick changes, the rest of the definition stays the same.

D. State representation

A state is a single fact: which location the student is standing at.

```
CurrentLocation = Administration Block
```

This gives exactly six possible states, one per location. I deliberately did not put the path taken so far, the distance walked so far, or the time of day inside the state. The reason is a useful test for any state design: a state should hold only what is needed to decide what can happen next. Where I can go from the Library is the same whether I came from the Gate or the Science Lab, so history does not belong in the state. History is tracked by the search algorithm in Weeks 2 and 3, not by the problem itself.

E. Actions

There is one kind of action: walk along a connection to a neighbouring location. Movement works in both directions, so every connection gives two actions, one each way. A move is legal only if a connection exists between the current location and the destination. You cannot jump from the Main Gate to the Science Lab directly, because no path joins them.

Legal actions from each state:

If the student is at	They can move to
Main Gate (L1)	Administration Block (cost 1), Cafeteria (cost 3)
Administration Block (L2)	Main Gate (1), Library (2), Science Lab (2)
Library (L3)	Administration Block (2)
Science Lab (L4)	Administration Block (2), Student Affairs (2)
Student Affairs (L5)	Cafeteria (2), Science Lab (2)
Cafeteria (L6)	Main Gate (3), Student Affairs (2)

F. Goal test

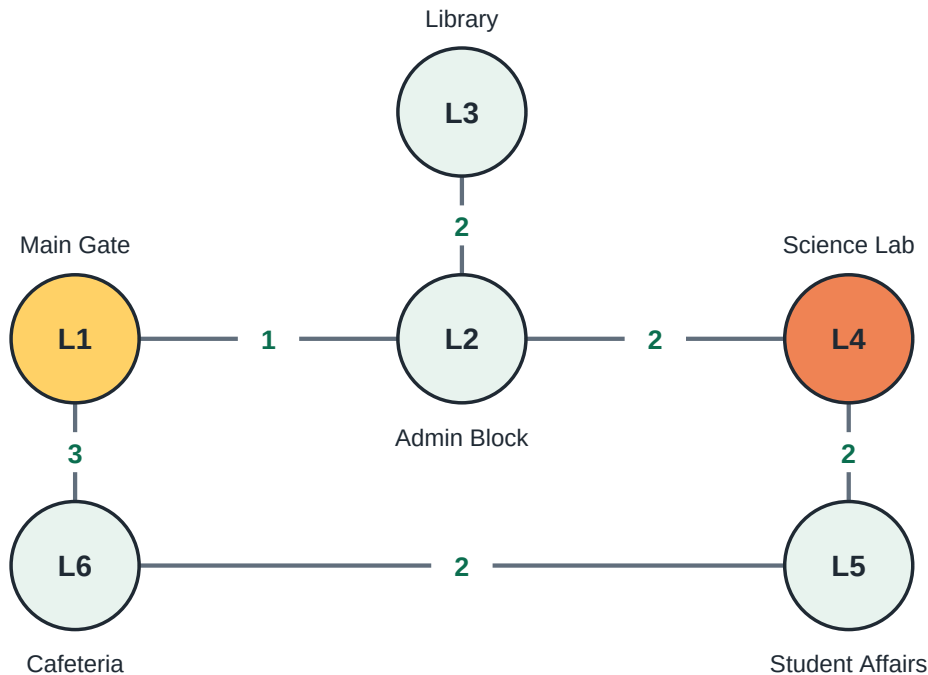
After every move the system asks one yes or no question: is `CurrentLocation` equal to the chosen goal? If yes, the search stops and the path that led here is the answer. If no, the search keeps going. It is a plain equality check, and it only works because the state is a single location. Nothing else has to be compared.

G. Cost

The number on a connection is the walking distance in kilometres. The cost of a whole route is the sum of the connections used. For example, Main Gate to Administration Block to Science Lab costs $1 + 2 = 3$ km. Every connection costs at least 1, so no move is ever free. That fact is what makes the hop-count heuristic safe in Week 3.

H. State-space graph

Each circle is a state and each line is a pair of actions (one per direction). The number on a line is its cost. The yellow circle is the initial state and the orange circle is the goal state I picked.



From	To	Cost (km)
Main Gate	Administration Block	1
Administration Block	Library	2
Administration Block	Science Lab	2
Main Gate	Cafeteria	3
Cafeteria	Student Affairs	2
Science Lab	Student Affairs	2

What the graph tells me before any code exists

It has 6 states and 6 connections. The Library is a dead end, since its only exit leads back to the Administration Block. The other five states form a loop: Main Gate, Administration Block, Science Lab, Student Affairs, Cafeteria, and back to Main Gate. Loops matter because a search that has no memory can walk around one forever, so in Week 2 the algorithms will need to remember where they have already been.

For Main Gate to Science Lab there are two sensible routes. Going through the Administration Block costs $1 + 2 = 3$ km. Going the long way round through the Cafeteria and Student Affairs costs $3 + 2 + 2 = 7$ km. The optimal answer is the 3 km route, and the later parts of the project will test whether each algorithm finds it.